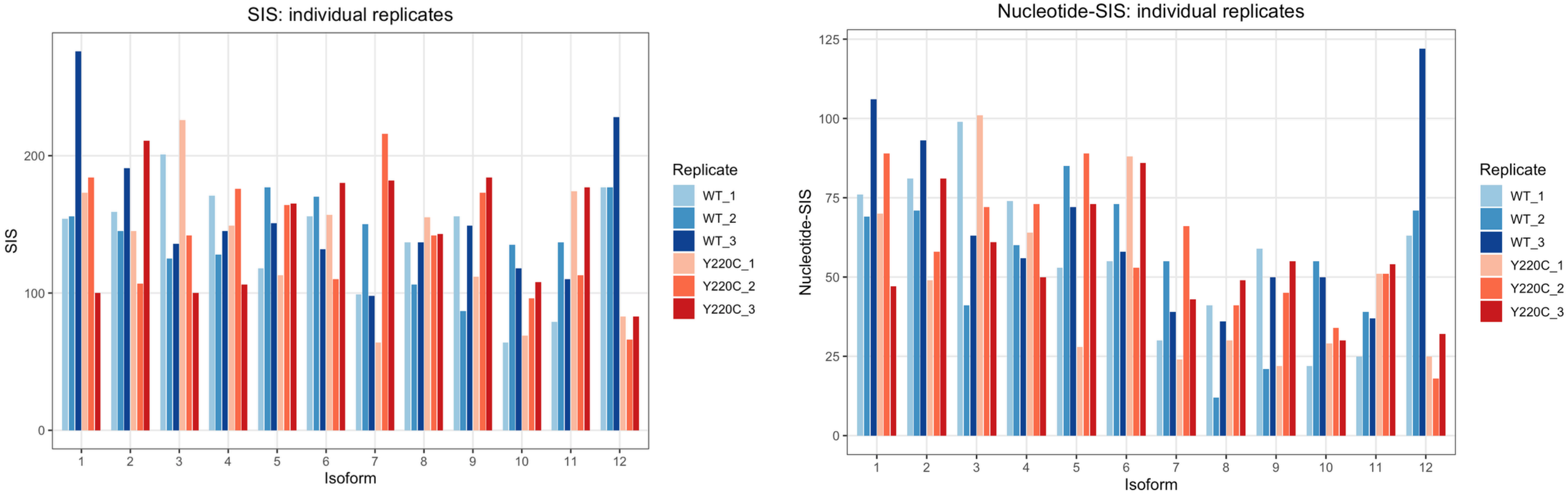
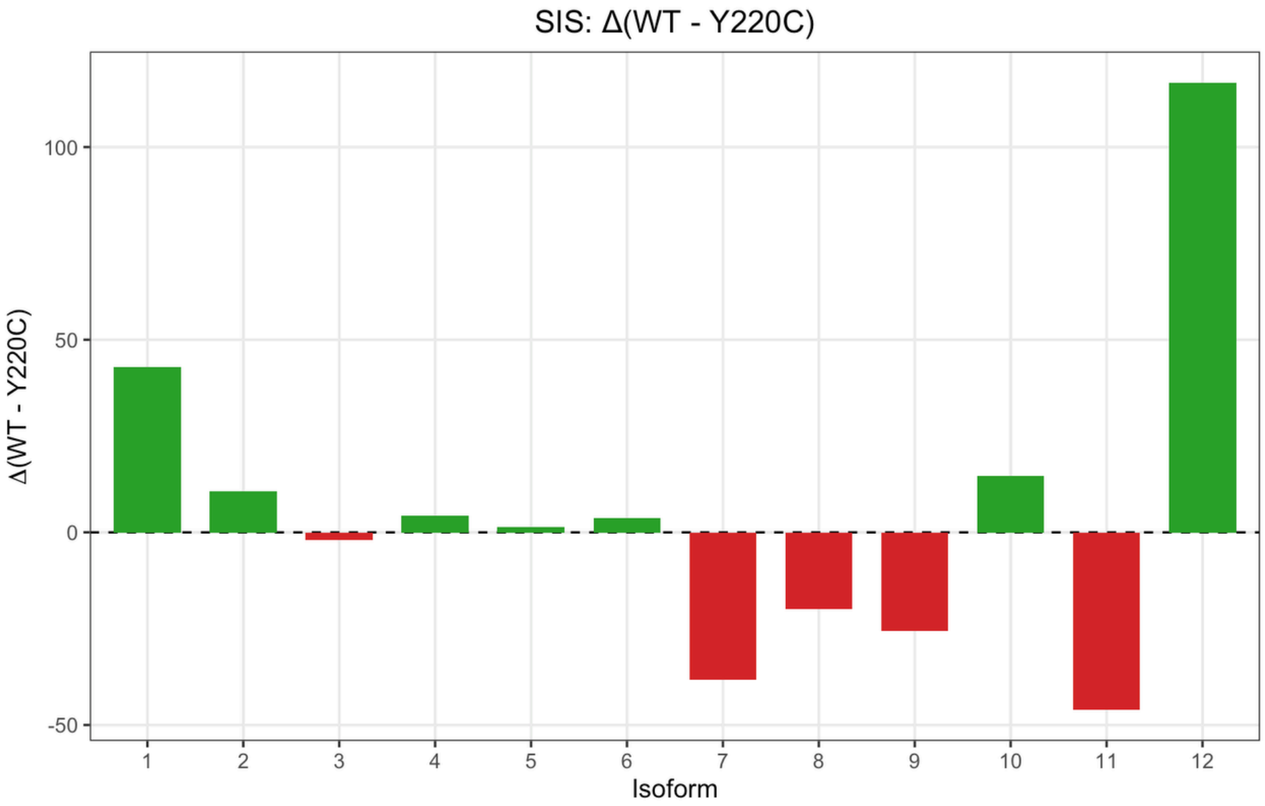


Figure S1. Replicate-level Specific Interaction Scores (SIS) and Nucleotide-Specific Interaction Scores across p53 isoforms. Bar plots show SIS (left) and Nucleotide-SIS (right) values for each of the three independent molecular dynamics replicates of wild-type (WT; blue shades) and Y220C (red shades) systems across all twelve p53 isoforms. While absolute values vary between replicates, the relative relationship between WT and Y220C remains consistent across most isoforms, with WT generally exhibiting higher specificity-weighted interaction scores.



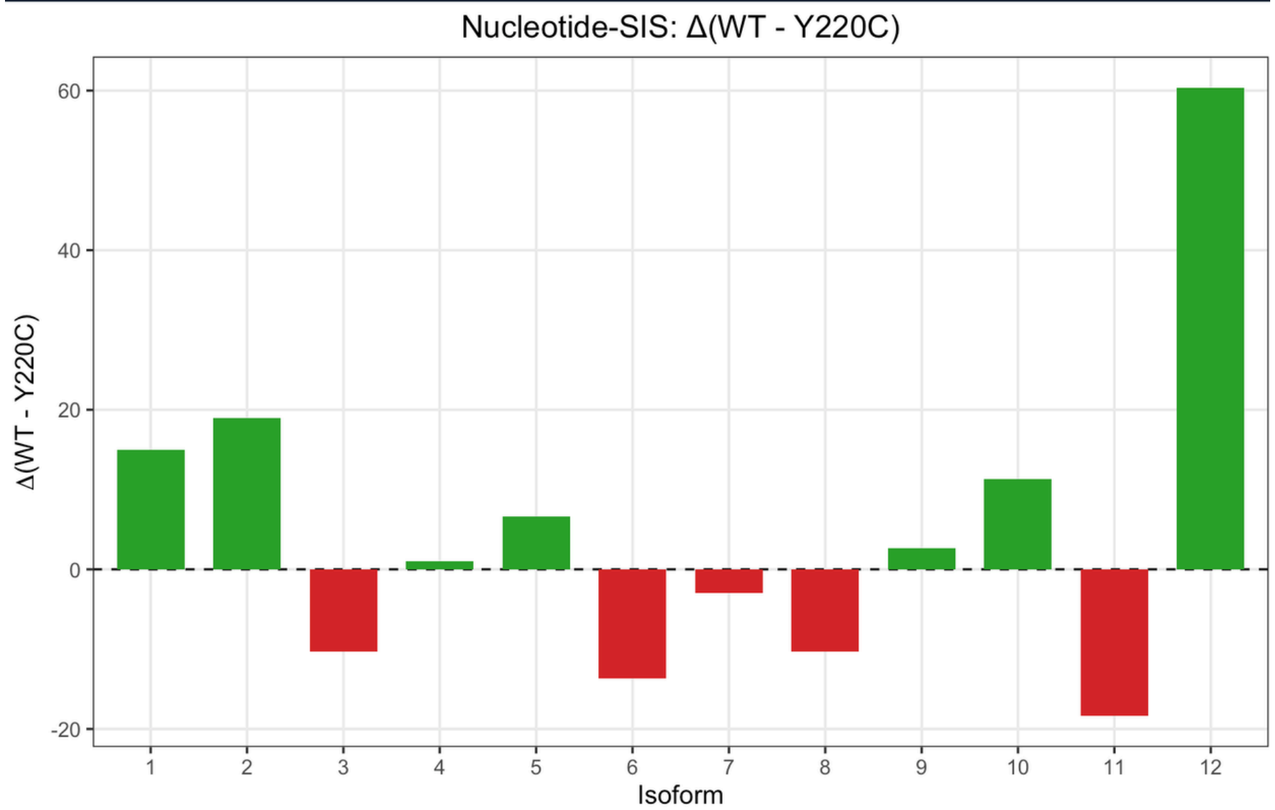


Figure S2. Isoform-resolved differences in DNA interactions between WT and Y220C p53.
Bar plots show the difference Δ(WT - Y220C) in Specific Interaction Score (SIS; left) and Nucleotide-Specific Interaction Score (right) for each of the twelve p53 isoforms. Positive values (green) indicate isoforms in which WT exhibits greater sequence-specific DNA interactions, whereas negative values (red) indicate relative enrichment in Y220C.

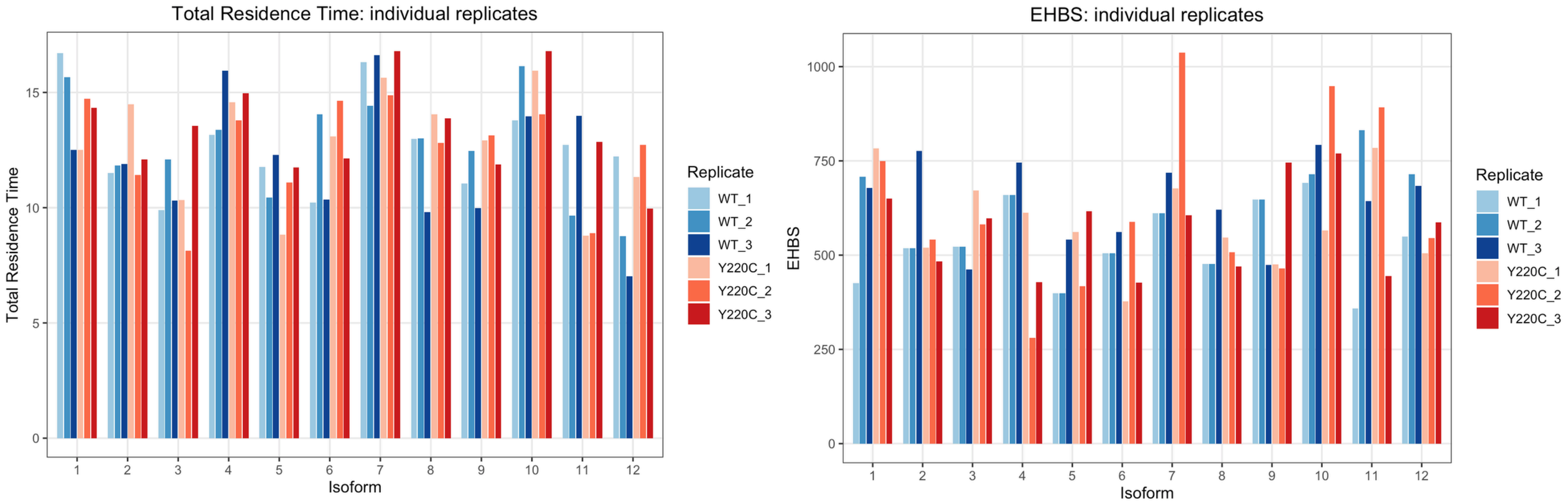


Figure S3. Replicate-level Total Residence Time (TRT) and Exhaustive Hydrogen Bond Sum (EHBS) across p53 isoforms. Bar plots show TRT (left) and EHBS (right) values for each of the three independent molecular dynamics replicates of wild-type (WT; blue shades) and Y220C (red shades) systems across all twelve isoforms. The relationship between WT and Y220C is consistent, with Y220C frequently exhibiting comparable or slightly elevated persistence and hydrogen-bond diversity metrics relative to WT. These replicate-resolved data support the averaged trends reported in the main text and demonstrate that the mutation-associated shift toward longer-lived but less specific DNA interactions is reproducible across independent trajectories.

Table S1. Reported Tissue Distribution of the Twelve Human p53 Isoforms

Isoform	Reported tissue and contexts.
p53α	Ubiquitous, all major tissues (brain, liver, lung, colon, immune cells, muscle)
p53β	Immune tissues (thymus, spleen, peripheral blood), epithelial tissues, colon
p53γ	Similar to β but generally lower levels; detected in immune cells and epithelial tissues
Δ40p53α	Brain, breast, liver, cell-cycle active tissues
Δ40p53β	Immune cells and some epithelial tissues (limited data)
Δ40p53γ	Low abundance; reported in cultured cells and some epithelial tissues
Δ133p53α	Colon, bone marrow, immune cells, tumors (especially colorectal, breast)
Δ133p53β	Immune tissues, inflammatory contexts, tumors
Δ133p53γ	Detected in cancers and some epithelial tissues
Δ160p53α	Tumor tissues (breast, melanoma, glioblastoma), stress conditions
Δ160p53β	Mostly cancer-associated detection
Δ160p53γ	Rare; reported in tumor samples

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